LAB 4

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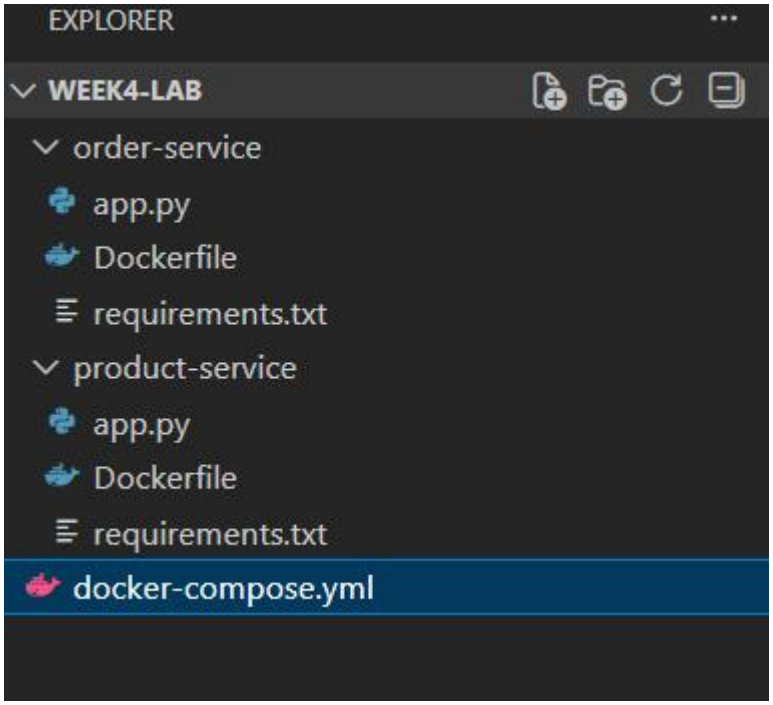
Group:SE2

PART A:

```
 Directory of C:\Users\hp\week4-lab

25/04/2026  23:47    <DIR>          .
25/04/2026  23:43    <DIR>          ..
25/04/2026  23:47                0 docker-compose.yml
25/04/2026  23:48    <DIR>          order-service
25/04/2026  23:48    <DIR>          product-service
               1 File(s)              0 bytes
               4 Dir(s)  78,997,606,400 bytes free

C:\Users\hp\week4-lab>dir product-service
 Volume in drive C has no label.
 Volume Serial Number is 908A-AE49

 Directory of C:\Users\hp\week4-lab\product-service

25/04/2026  23:48    <DIR>          .
25/04/2026  23:47    <DIR>          ..
25/04/2026  23:47                0 app.py
25/04/2026  23:48                0 Dockerfile
25/04/2026  23:48                0 requirements.txt
               3 File(s)              0 bytes
               2 Dir(s)  78,997,671,936 bytes free

C:\Users\hp\week4-lab>dir order-service
 Volume in drive C has no label.
 Volume Serial Number is 908A-AE49

 Directory of C:\Users\hp\week4-lab\order-service

25/04/2026  23:48    <DIR>          .
25/04/2026  23:47    <DIR>          ..
25/04/2026  23:48                0 app.py
25/04/2026  23:48                0 Dockerfile
25/04/2026  23:48                0 requirements.txt
               3 File(s)              0 bytes
               2 Dir(s)  78,997,594,112 bytes free
```



PART B:

product-service > app.py > ...

```python
from flask import Flask, jsonify

app = Flask(__name__)

PRODUCTS = {
    1: {"id": 1, "name": "Laptop", "price": 1200},
    2: {"id": 2, "name": "Phone", "price": 650},
    3: {"id": 3, "name": "Headphones", "price": 120}
}

@app.route("/health", methods=["GET"])
def health():
    return jsonify({"service": "product-service", "status": "up"}), 200

@app.route("/products/<int:product_id>", methods=["GET"])
def get_product(product_id):
    product = PRODUCTS.get(product_id)
    if product:
        return jsonify(product), 200
    return jsonify({"error": "Product not found"}), 404

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5001)
```

PART C:

order-service > app.py > ...

```python
import os
import time
import requests
from flask import Flask, jsonify, request

app = Flask(__name__)

PRODUCT_SERVICE_URL = os.getenv("PRODUCT_SERVICE_URL", "http://product-service:5001")

def fetch_product(product_id, retries=2, delay=1):
    url = f"{PRODUCT_SERVICE_URL}/products/{product_id}"
    for attempt in range(retries + 1):
        try:
            response = requests.get(url, timeout=2)
            return response
        except requests.exceptions.RequestException:
            if attempt < retries:
                time.sleep(delay)
            else:
                return None

@app.route("/health", methods=["GET"])
def health():
    return jsonify({"service": "order-service", "status": "up"}), 200

@app.route("/orders", methods=["POST"])
def create_order():
    data = request.get_json()
    product_id = data.get("product_id")
    quantity = data.get("quantity", 1)

    response = fetch_product(product_id)

    if response is None:
        return jsonify({"error": "product-service unavailable"}), 503

    if response.status_code != 200:
        return jsonify({"error": "invalid product"}), 400

    product = response.json()
    total = product["price"] * quantity

    return jsonify({
        "message": "Order created",
        "product": product["name"],
        "quantity": quantity,
        "total_price": total
    }), 201

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5002)
```

PART D:

product-service/Dockerfile:

product-service > Dockerfile > ...

```
FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

EXPOSE 5001

CMD ["python", "app.py"]
```

order-service/Dockerfile:

order-service > Dockerfile > ...

```
FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

EXPOSE 5002

CMD ["python", "app.py"]
```

PART E

File Edit Selection View Go ... week4-lab

EXPLORER

WEEK4-LAB
- order-service
  - app.py
  - Dockerfile
  - requirements.txt
- product-service
  - app.py
  - Dockerfile
  - requirements.txt
- docker-compose.yml

requirements.txt product-service | app.py order-service | Dockerfile product-service | Dockerfile order-service | docker-compose.yml | app.py product-service

docker-compose.yml

```yaml
version: "3.9"

services:
  product-service:
    build: ./product-service
    container_name: product-service
    ports:
      - "5001:5001"
    healthcheck:
      test: ["CMD", "python", "-c", "import urllib.request; urllib.request.urlopen('http://localhost:5001/health')"]
      interval: 10s
      timeout: 5s
      retries: 3
    restart: unless-stopped

  order-service:
    build: ./order-service
    container_name: order-service
    ports:
      - "5002:5002"
    environment:
      PRODUCT_SERVICE_URL: http://product-service:5001
    depends_on:
      - product-service
    healthcheck:
      test: ["CMD", "python", "-c", "import urllib.request; urllib.request.urlopen('http://localhost:5002/health')"]
      interval: 10s
      timeout: 5s
      retries: 3
    restart: unless-stopped
```

PART F:

```
C:\Users\hp\week4-lab>docker compose up --build -d
time="2026-04-26T00:02:59+03:00" level=warning msg="C:\\Users\\hp\\week4-lab\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential co
#1 [internal] load local bake definitions
#1 reading from stdin 1.05kB done
#1 DONE 0.0s

#2 [product-service internal] load build definition from Dockerfile
#2 transferring dockerfile: 217B 0.0s done
#2 DONE 0.1s

#3 [order-service internal] load build definition from Dockerfile
#3 transferring dockerfile: 217B 0.0s done
#3 DONE 0.1s

#4 [product-service internal] load metadata for docker.io/library/python:3.12-slim
#4 DONE 1.1s

#5 [order-service internal] load .dockerignore
#5 transferring context: 2B done
#5 DONE 0.0s

#6 [product-service internal] load .dockerignore
#6 transferring context: 2B done
#6 DONE 0.0s

#7 [product-service internal] load build context
#7 transferring context: 768B 0.0s done
#7 DONE 0.1s

#8 [product-service 1/5] FROM docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e4590c18c2e3
#8 resolve docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e4590c18c2e3 0.0s done
#8 DONE 0.1s

#9 [order-service internal] load build context
#9 transferring context: 1.59kB 0.0s done
#9 DONE 0.1s

#10 [product-service 2/5] WORKDIR /app
#10 CACHED

#11 [product-service 3/5] COPY requirements.txt .
#11 DONE 0.1s

#12 [order-service 3/5] COPY requirements.txt .
#12 DONE 0.1s

#13 [order-service 4/5] RUN pip install --no-cache-dir -r requirements.txt
#13 3.241 Collecting flask==3.0.3 (from -r requirements.txt (line 1))
```

```
#17 [order-service 5/5] COPY app.py .
#17 DONE 0.1s

#18 [order-service] exporting to image
#18 ...

#16 [product-service] exporting to image
#16 exporting layers 0.9s done
#16 exporting manifest sha256:3c223971c9b6cc05f436b0b084918490fb686655965b92ce4ebda3bbd40144ed 0.0s done
#16 exporting config sha256:c45eae149321da7cd8ab54f64d2e72a499c74712335b08aa0e229052fd68d657 0.0s done
#16 exporting attestation manifest sha256:4552acd8796f0086748d293e5c2f2f44728b7c466c56f08c516dd92c99bdfcb0 0.0s done
#16 exporting manifest list sha256:46aee317b6f8331a6c1fa59473f8044197b4df8317157a6eab380b3ac86da39e 0.0s done
#16 naming to docker.io/library/week4-lab-product-service:latest done
#16 unpacking to docker.io/library/week4-lab-product-service:latest 0.6s done
#16 DONE 1.6s

#19 [product-service] resolving provenance for metadata file
#19 DONE 0.0s

#18 [order-service] exporting to image
#18 exporting layers 1.1s done
#18 exporting manifest sha256:9b1719ad5545939e1c1fa4df460d74c96544a1b9b81efc4f03d1e964e10a8b27 0.0s done
#18 exporting config sha256:6e818be2f18cc7a2779ba8132d7735cfd41e31c3d080ba24863a05a7d8b1479c 0.0s done
#18 exporting attestation manifest sha256:8b8e0a0e8dae4ba192c1aa69fe479344705e25e5c3dd229b60ac70138aa85db9 0.0s done
#18 exporting manifest list sha256:dc6ec5c1416960818dd082da036f772507be3bb785f2a10db3d36fb741ca02a6
#18 exporting manifest list sha256:dc6ec5c1416960818dd082da036f772507be3bb785f2a10db3d36fb741ca02a6 0.0s done
#18 naming to docker.io/library/week4-lab-order-service:latest done
#18 unpacking to docker.io/library/week4-lab-order-service:latest
#18 unpacking to docker.io/library/week4-lab-order-service:latest 0.6s done
#18 DONE 1.9s

#20 [order-service] resolving provenance for metadata file
#20 DONE 0.0s
[+] up 5/5
 ✔Image week4-lab-order-service   Built                                                                    12.8s
 ✔Image week4-lab-product-service Built                                                                    12.8s
 ✔Network week4-lab_default       Created                                                                   0.1s
 ✔Container product-service       Started                                                                   1.2s
 ✔Container order-service         Started                                                                   0.8s

C:\Users\hp\week4-lab>docker compose ps
time="2026-04-26T00:07:12+03:00" level=warning msg="C:\\Users\\hp\\week4-lab\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME             IMAGE                    COMMAND            SERVICE           CREATED         STATUS                   PORTS
order-service    week4-lab-order-service  "python app.py"    order-service     3 minutes ago   Up 3 minutes (healthy)   0.0.0.0:5002->5002/tcp, [::]:5002->5002/tcp
product-service  week4-lab-product-service "python app.py"   product-service   4 minutes ago   Up 3 minutes (healthy)   0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp

C:\Users\hp\week4-lab>
```

PART G:

```
C:\Users\hp\week4-lab>curl http://localhost:5001/health
{"service":"product-service","status":"up"}

C:\Users\hp\week4-lab>curl http://localhost:5001/products/1
{"id":1,"name":"Laptop","price":1200}

C:\Users\hp\week4-lab>curl http://localhost:5001/products/99
{"error":"Product not found"}

C:\Users\hp\week4-lab>curl -X POST http://localhost:5002/orders -H "Content-Type: application/json" -d "{\"product_id\": 1, \"quantity\": 2}"
{"message":"Order created","product":"Laptop","quantity":2,"total_price":2400}

C:\Users\hp\week4-lab>
```

PART H:

```
C:\Users\hp\week4-lab>docker stop product-service
product-service

C:\Users\hp\week4-lab>curl -X POST http://localhost:5002/orders -H "Content-Type: application/json" -d "{\"product_id\": 2, \"quantity\": 1}"
{"error":"product-service unavailable"}

C:\Users\hp\week4-lab>
```

Restart service

```
C:\Users\hp\week4-lab>curl -X POST http://localhost:5002/orders -H "Content-Type: application/json" -d "{\"product_id\": 2, \"quantity\": 1}"
{"error":"product-service unavailable"}

C:\Users\hp\week4-lab>docker start product-service
product-service

C:\Users\hp\week4-lab>docker compose ps
time="2026-04-26T01:00:31+03:00" level=warning msg="C:\\Users\\hp\\week4-lab\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME              IMAGE                     COMMAND           SERVICE           CREATED          STATUS                        PORTS
order-service     week4-lab-order-service   "python app.py"   order-service     57 minutes ago   Up 57 minutes (healthy)       0.0.0.0:5002->5002/tcp, [::]:5002->5002/tcp
product-service   week4-lab-product-service "python app.py"   product-service   57 minutes ago   Up 8 seconds (health: starting)   0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp

C:\Users\hp\week4-lab>
```

Part I:

1- Microservices vs Monolith

Each service is independent (product & order)

Easier to scale individual services

Fault isolation (one service can fail without crashing entire system)

Independent deployment using Docker

2. New Complexities Introduced

Service communication over network

Dependency issues : order-service depends on product-service

More configuration (Docker, environment variables) harder debugging compared to monolith

3. order-service becomes slow,requests timeout,user gets delayed or failed responses

cascading failures can occur

4. 12-Factor Principles Used

Config via environment variables (PRODUCT_SERVICE_URL)

Stateless services,Port binding (5001, 5002),Logs via stdout (Docker logs),Service independence